

FNU SWATI

☎ 646-729-6173 📍 New York, NY ✉ swati.guleria2319@gmail.com [in LinkedIn](#) [GitHub](#)

EDUCATION

Master of Science in Data Science (Computer Science)

Stony Brook University(SUNY), New York

May 2026

GPA: 3.40/4.00

Bachelor of Technology in Computer Science

Chitkara Institute of Science and Technology, India

July 2019

GPA: 3.67/4.00

EXPERIENCE

Research Assistant

May 2026 – Present

Stony Brook University

New York, NY

- Designed and implemented an LLM simulation pipeline in Python to evaluate whether **GPT**, **Claude**, and **Centaur** can replicate human survey responses across 5 persona-conditioning conditions, using SQLite for experiment tracking and the Anthropic/OpenAI APIs for model inference.
- Conducted **causal inference** research benchmarking LLM algorithmic fidelity against **144 human respondents** using MAE, convex hull consistency tests, and stakeholder alignment metrics to assess in-distribution vs. out-of-distribution generalization of large language models.

AI Software Engineer Intern

May 2025 – Aug 2025

Voltihost LLC

New York, NY

- Delivered an ML demand forecasting pipeline with **15+** engineered features achieving **91%** accuracy, serving 7 to 30 day predictions in sub-3s latency via **ONNX** runtime on **Vercel** serverless infrastructure
- Built an agentic **text-to-SQL** system for a demand analytics tool using GPT-4o function calling and schema-aware prompting, enabling sales teams to query forecasting data with **92%** SQL generation accuracy

Senior Software Engineer (ML)

Jan 2022 – May 2024

Infosys

Pune, India

- Developed an IT support ticket classification system for Finacle e-banking using fine-tuned **DistilBERT** and MLflow experiment tracking, deployed on AWS SageMaker with CodePipeline/CodeBuild, achieving **85%** accuracy that halved SLA response times and saved **\$0.6M** annually
- Enhanced the support pipeline with **QLoRA** fine-tuned **Llama 2** for domain-specific summarization and a retrieval layer using LangChain and FAISS, enabling auto-routing and contextual resolution suggestions that reduced manual ticket triage by **~35%**
- Implemented a REST API-enabled ID document classification service using **AWS Textract** and Lambda, with unit-tested serverless functions handling extraction and routing across multiple Finacle banking workflows

Software Engineer

Aug 2019 – Dec 2021

Infosys | Awards: Insta Award, Constant Contributor Badge

Pune, India

- Led a 3-member DevOps team owning CIT, E2E, and production environments for **50+** integrations across EMEA, SEPA, and APAC for Bank of America, supporting **\$3M** in annual client revenue
- Built and orchestrated Dockerized applications on AWS EKS using Kubernetes, JFrog, and ECR, creating **15+** CI/CD pipelines with GitLab and Ansible that cut deployment time by **50%**
- Provisioned Finacle Core Banking on OpenShift and AWS clusters, standing up and managing environments for **20+** applications with **5+** installations annually to support evolving business requirements
- Maintained production reliability across **10+** cloud servers through **cron automation**, log archiving, and disk management, while replicating 50+ databases across accounts to support application structure changes in APAC

PROJECTS

Cross-Lingual Document Retrieval using Embedding Models | Python, Docker, LangChain, Weaviate, Qwen 2.5 VLM | [Link](#)

- Designed a cross-lingual RAG pipeline enabling semantic search across **5 languages** over **1,000+** indexed multilingual documents, achieving 89% retrieval accuracy (MRR) on topic matching and **81%** on mono-lingual retrieval
- Benchmarked **3** embedding models across **50** cross-lingual queries, identifying the top performer with **4%** higher MRR, deployed as a fully local privacy-preserving search system with an interactive chat interface via **Open WebUI**

CustIQ360°: Customer Intelligence Platform | Gemini 2.5 Flash, LangGraph, RAG, FAISS, Gemini Vision AI | [Link](#)

- Architected a multi-agent AI banking dashboard for Relationship Managers using a 5-node LangGraph agentic pipeline with SSE streaming, reducing manual customer research time by **60%** across **1K+** synthetic customer records.
- Designed an AI powered **KYC document extraction** pipeline supporting **5+** document types across **APAC/EMEA** regions, with auto-persist to customer profiles and multi-currency Financial Simulator(EMI, FD, Loan), cutting manual effort by **70%**.

SKILLS

- Programming Languages:** Python, C++, SQL/PL-SQL, Bash, Shell Scripting, Linux , React, Angular
- Machine Learning:** Pandas, NumPy, Scikit-Learn, PyTorch, HuggingFace, LangChain, FAISS
- Cloud & MLOps:** vLLM, Vercel, Kubernetes, Docker, GitHub Actions, GitLab, Ansible, RDS , AWS (EKS, EC2, S3, SageMaker, IAM, Lambda), MLflow, Airflow, OpenShift, Jenkins, CodePipeline , Jira, RabbitMQ , Terraform , SVN, Debugging, Grafana
- Databases:** Weaviate, Qdrant, Oracle 19c , Supabase, PostgreSQL, MySQL
- AI tools:** Claude Code, Cursor, Github Copilot